arXiv:2410.19714v3 [math.CO] 18 Feb 2026

# Garsia–Remmel $q$-rook numbers are not always unimodal

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September 2025

*Dedicated to the memory of Adriano Garsia.*

**Abstract**

We show by an explicit example that the Garsia–Remmel $q$-rook numbers of Ferrers boards do not all have unimodal sequences of coefficients. This resolves in the negative a question from 1986 by the aforementioned authors.

**keywords:** $q$-rook numbers, rook placements, Ferrers boards, unimodality, counterexamples, q-Stirling numbers

The central objects in rook theory are the **placements of (nonattacking) rooks** on a **board**. Formally, a board $B$ is a finite subset of $\mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$, and a rook placement on $B$ is a subset $w$ of $B$ such that no two elements of $w$ agree on either coordinate. The terminology comes from the natural visual representation of these objects, as in Figure 1. The **rook number** $R_r(B)$ is defined as the number of placements of $r$ rooks on the board $B$.

A case of particular interest (beginning in the first systematic study [KR46] of the field) is when $B = B_\lambda$ is a **Ferrers board** associated to an **integer partition** $\lambda$: that is, $\lambda = \langle \lambda_1, \ldots, \lambda_\ell \rangle$ is a finite nonincreasing sequence of positive integers and $B_\lambda := \{(i,j) : 1 \leq i \leq \ell, 1 \leq j \leq \lambda_i\}$. For example, in the case of the **staircase partition** $\delta_n := \langle n-1, n-2, \ldots, 2, 1 \rangle$, one has that

$$R_k(B_{\delta_n}) = S(n, n-k)$$

is a Stirling number of the second kind [KR46, §5]. In general, two partitions $\lambda, \mu$ are said to be **rook equivalent** if $R_k(B_\lambda) = R_k(B_\mu)$ for all $k$ [GJW75]. This condition can be phrased

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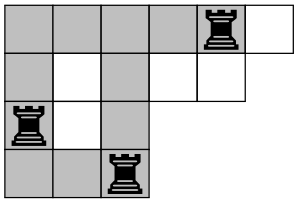


Figure 1: The rook placement $\{(1,5),(3,1),(4,3)\}$ on the board $B_{\langle 6,5,3,3\rangle}$ has inversion number 5. (Here we use matrix coordinates.)

in various ways; for our purposes, the most convenient characterization [CGR23, Cor. 3.2] is that $\lambda$ and $\mu$ are rook equivalent if and only if $d_m(\lambda) = d_m(\mu)$ for all $m$, where the $d_m$ are the **diagonal counts**

$$d_m(\nu) := |\{(i,j) \in B_\nu : i + j = m + 1\}|. \tag{1}$$

In [GR86], Garsia and Remmel introduced a $q$-analogue of the rook number. This analogue is in terms of the **inversion number**, defined as follows: given a rook placement $w$ on a Ferrers board $B_\lambda$, for each rook $(a,b) \in w$, delete from $B_\lambda$ all elements $(i,b)$ with $i \le a$ and $(a,j)$ with $j \le b$; the inversion number $\mathrm{inv}(w)$ is the number of undeleted cells. Then the **(Garsia–Remmel) $q$-rook number** of **rank** $k$ is

$$R_k(\lambda; q) := \sum_{\substack{w \text{ a placement of} \\ k \text{ rooks on } B_\lambda}} q^{\mathrm{inv}(w)}.$$

By definition, $R_k(\lambda; q)$ is a polynomial in the formal variable $q$ with nonnegative integer coefficients such that $R_k(\lambda; 1) = R_k(B_\lambda)$. For example, the rank-3 $q$-rook number of the partition $\langle 6,5,3,3\rangle$ is $q^{11} + 4q^{10} + 10q^9 + 19q^8 + 27q^7 + 30q^6 + 25q^5 + 15q^4 + 6q^3 + q^2$, and the rook placement in Figure 1 is one of the 25 that contribute to the coefficient of $q^5$.

Analogously to the case $q = 1$, one says two partitions are **$q$-rook equivalent** if they have the same $q$-rook numbers. Surprisingly, two partitions are rook equivalent if and only if they are $q$-rook equivalent [GR86, p. 257]. In the particular case $\lambda = \delta_n$, the polynomial $R_k(\lambda; q)$ is a *$q$-Stirling number of the second kind* [GR86, Eq. I.9]; for more on these numbers, see [Mil82; WW91] and Remarks 8 and 10 below.

The same polynomials $R_k(\lambda; q)$ also arise in enumerative linear algebra, when counting matrices of rank $k$ over a finite field whose nonzero entries all belong to $B_\lambda$ [Hag98, Thm. 1]; this connection has led to applications in coding theory, since up to a power of $q$, the $R_k(\lambda; q)$ give the weights of certain rank-metric codes (e.g., see [Rav18; GR20; GR24]). The $q$-rook numbers and the related *$q$-hit numbers* have also recently appeared in the expansions of *chromatic symmetric functions* of graphs corresponding to certain Dyck paths called *abelian* [AN21; CMP23; NT21; HOY24].

Based on an assortment of empirical and circumstantial evidence, Garsia and Remmel conjectured [GR86, p. 250] that the polynomials $R_k(\lambda; q)$ always have **unimodal sequence of coefficients**, i.e., that if

$$R_k(\lambda; q) = a_N q^N + a_{N-1} q^{N-1} + \ldots + a_1 q + a_0$$

then for some $m$ it holds that $a_N \leq a_{N-1} \leq \cdots \leq a_{m+1} \leq a_m \geq a_{m-1} \geq \cdots \geq a_1 \geq a_0$. As we now show, it is not difficult to prove the Garsia–Remmel unimodality conjecture for the cases of the maximum and minimum nontrivial numbers of rooks.

**Proposition 1.** *For every partition $\lambda = \langle \lambda_1, \ldots, \lambda_\ell \rangle$, $R_\ell(\lambda; q)$ is unimodal.*

*Proof.* In the case of $\ell$ rooks, there must be one rook in each row. Consider iteratively placing the rooks beginning with the last row. Regardless of how one has placed rooks in rows $k+1, \ldots, \ell$, there will be $\lambda_k - (\ell - k)$ legal positions to place a rook in row $k$, and the different choices will leave $0, 1, \ldots, \lambda_k - (\ell - k) - 1$ inversions in row $k$. Consequently

$$R_\ell(\lambda; q) = [\lambda_\ell]_q \cdot [\lambda_{\ell-1} - 1]_q \cdots [\lambda_1 - (\ell - 1)]_q \tag{2}$$

where $[n]_q := 1 + q + q^2 + \ldots + q^{n-1}$. Thus $R_\ell(\lambda; q)$ is a product of log-concave polynomials (i.e., polynomials for which the coefficient sequence satisfies $a_i^2 \geq a_{i-1}a_{i+1}$) with nonnegative coefficients and no internal zeros. The family of such polynomials is closed under products [Sta89, Prop. 2], and furthermore every such polynomial is unimodal [Sta89, p. 500]. □

**Proposition 2.** *For every partition $\lambda$, $R_1(\lambda; q)$ is unimodal.*

*Proof.* It is easy to see [CGR23, proof of Cor. 3.2] that

$$R_1(\lambda; q) = \sum_i d_i(\lambda) \cdot q^{|\lambda| - i} \tag{3}$$

where $d_i$ is as in (1) and $|\lambda| = \lambda_1 + \lambda_2 + \cdots + \lambda_\ell$ is the **size** of $\lambda$. This polynomial is unimodal because, taking $m$ to be the maximum index for which $d_m = m$ and fixing a choice of a box $(a, m + 2 - a)$ that does *not* belong to the $(m+1)$st diagonal of $B_\lambda$, we have that the map

$$(x, y) \mapsto \begin{cases} (x, y-1) & \text{if } x < a, \\ (x-1, y) & \text{if } x > a \end{cases}$$

is an injection from the $k$th diagonal of $B_\lambda$ to the $(k-1)$st for all $k > m$, and hence $d_m \geq d_{m+1} \geq \cdots$. □

Progress on the general conjecture has been limited. One reason for this is that, although $q$-rook numbers of maximum rank are always log-concave (per the proof of Proposition 1), this is not the case for lower-rank $q$-rook numbers, as shown in the next example. Thus, the assorted machinery that has been built up (especially in the past decade [AHK18; BH20; AOV21; CP24]) to prove log-concavity results is not directly useful here.

**Example 3.** One has (either from (3) or by direct examination) that

$$R_1(\langle 4, 1 \rangle; q) = q^4 + 2q^3 + q^2 + q,$$

which is not log-concave (since $a_2^2 = 1 < 2 = a_1 \cdot a_3$).

A second reason that progress on the general conjecture has been limited is that the conjecture is false.

**Proposition 4.** *The $q$-rook number*

$$R_2(\langle 10, 9, 3, 2, 1\rangle; q)$$

*is not unimodal.*

*Proof.* By considering whether or not the cell $(\ell, \lambda_\ell)$ has a rook, it is easy to see that the $q$-rook numbers satisfy the recurrence relation

$$R_k(\langle \lambda_1, \ldots, \lambda_{\ell-1}, \lambda_\ell\rangle; q) = q \cdot R_k(\langle \lambda_1, \ldots, \lambda_{\ell-1}, \lambda_\ell - 1\rangle; q) + R_{k-1}(\langle \lambda_1 - 1, \ldots, \lambda_{\ell-1} - 1\rangle; q). \quad (4)$$

(This is a special case of Dworkin's "deletion-contraction" recurrence for $q$-rook numbers [Dwo98, Thm. 6.10].) Repeated applications of (4) yield

$$R_2(\langle 10, 9, 3, 2, 1\rangle; q) = q \cdot R_2(\langle 10, 9, 3, 2\rangle; q) + R_1(\langle 9, 8, 2, 1\rangle; q),$$
$$R_2(\langle 10, 9, 3, 2\rangle; q) = q^2 \cdot R_2(\langle 10, 9, 3\rangle; q) + (1 + q) \cdot R_1(\langle 9, 8, 2\rangle; q),$$

and

$$R_2(\langle 10, 9, 3\rangle; q) = q^3 \cdot R_2(\langle 10, 9\rangle; q) + (1 + q + q^2) \cdot R_1(\langle 9, 8\rangle; q).$$

By (2), we have $R_2(\langle 10, 9\rangle; q) = ([9]_q)^2$, while the various instances of $R_1$ can be computed from (3). Plugging these values in and simplifying produces

$$R_2(\langle 10, 9, 3, 2, 1\rangle; q) = q^{22} + 3q^{21} + 7q^{20} + 13q^{19} + 18q^{18} + 21q^{17} + 22q^{16} + 20q^{15} + 21q^{14} + 20q^{13} + 17q^{12} + 12q^{11} + 5q^{10} + 4q^9 + 3q^8 + 2q^7 + q^6,$$

which is manifestly non-unimodal. □

**Remark 5.** The example in Proposition 4 is the smallest partition that has a non-unimodal $q$-rook number: we checked by exhaustive computation (carried out in `Sage` [The24], implemented with the simple recursion (4), with base case $R_0(\lambda; q) = q^{|\lambda|}$) that if $|\lambda| \leq 24$, then the $q$-rook number $R_k(\lambda; q)$ is unimodal for all $k$. The only other partitions of size 25 with a non-unimodal $q$-rook number are rook equivalent to $\langle 10, 9, 3, 2, 1\rangle$. For larger $n$, the number of partitions of $n$ with non-unimodal $q$-rook number in rank 2 grows steadily; for example, there are 67 rook equivalence classes of partitions of size 40 with $R_2$ non-unimodal (out of 1113 total classes).

**Remark 6.** We record here the results of some further computational experiments:

- The smallest partitions $\lambda$ for which $R_3(\lambda; q)$ is not unimodal are $\langle 14, 13, 5, 4, 3, 2, 1\rangle$, of size 42, and the partitions rook equivalent to it. These partitions are also non-unimodal in rank 2.
- The partitions $\lambda$ of size 43 for which $R_3(\lambda; q)$ is not unimodal are $\langle 16, 12, 5, 4, 3, 2, 1\rangle$ and the partitions rook equivalent to it. These partitions *are* unimodal in rank 2.

- The smallest partitions $\lambda$ for which $R_4(\lambda;q)$ is not unimodal are $\langle 31,17,16,5,4,3,2\rangle$, of size 78, and the partitions rook equivalent to it. These partitions are also non-unimodal in rank 2 but are unimodal in rank 3.

- For the partition $\langle 69,67,65,10,9,7,6,5,4,3,2\rangle$ of size 247, $R_5(\lambda;q)$ is not unimodal. This partition was found doing an exploratory search with code written with the help of `Gemini` [Goo26]. We have no reason to believe this is a minimum-size example, but we have not found a smaller one. See Table 1 for similar non-unimodal examples for $k=6,7,8,9,10$. The code for these explorations is available on the `arXiv` as an ancillary file and a website with a visualization tool for $q$-rook numbers is available at `https://ahmorales.math.uqam.ca/Animations/qrooks.html`.

| $k$ | Partition $\lambda$ | size |
|---|---|---|
| 2 | $\langle 10,9,3,2,1\rangle$ | 25 |
| 3 | $\langle 14,13,5,4,3,2,1\rangle$ | 42 |
| 4 | $\langle 31,17,16,5,4,3,2\rangle$ | 78 |
| 5 | $\langle 69,67,65,10,9,7,6,5,4,3,2\rangle$ | 247 |
| 6 | $\langle 67^2,14^{11},13,12,11,10,9,8,7,6,5,4,3,2,1\rangle$ | 379 |
| 7 | $\langle 78^2,19^4,18^6,17,16,15,14,13,12,11,10,9,8,7,6,5,4,3,2,1\rangle$ | 494 |
| 8 | $\langle 122,\ 119,\ 31,\ 30,\ 29,\ 28,\ 27,\ 26,\ 25,\ 24,\ 23,\ 22,\ 21,\ 20,\ 19,\ 18,$ $17,\ 16,\ 15,\ 14,\ 13,\ 12,\ 11,\ 10,\ 9,\ 8,\ 7,\ 6,\ 5,\ 4,\ 3,\ 2,\ 1\rangle$ | 785 |
| 9 | $\langle 216^2,39,38^2,37,34^4,33^2,32^2,30^4,29,28,24^2,22,20^2,19,18,$ $15^4,13^2,1^4\rangle$ | 1176 |
| 10 | $\langle 244^2,66,65,62,59,57^3,56^4,54,43,42^2,40,39,35,34^2,21^3,20^5,$ $18,16^2,13^2,12^4,9^5,1^{14}\rangle$ | 1801 |

Table 1: Examples of partitions $\lambda$ such that $R_k(\lambda;q)$ is not unimodal for $k=2,\ldots,10$. The examples for $k=2,3,4$ are minimum-size examples, while those for $k=5,\ldots,10$ were found doing a exploratory search with code written with the help of `Gemini` [Goo26] based on the heuristic of `PatternBoost` [Cha+24].

It seems reasonable to conjecture that for each integer $k>1$, there is some partition $\lambda$ for which $R_k(\lambda;q)$ is not unimodal. Typically, it seems that when $R_k(\lambda;q)$ is not unimodal, there is a gap between $k$ and the number of parts $\ell(\lambda)$ of $\lambda$. For example, every partition $\lambda$ with less than 100 boxes for which $R_2(\lambda;q)$ is non-unimodal has $\ell(\lambda)\geq 5$. The smallest example with a gap of size 2 is for the non-unimodal polynomial $R_3(\langle 24,15,14,3,2\rangle;q)$. Is $R_{\ell-1}(\lambda;q)$ always unimodal?

**Remark 7.** In the tables of data they collected in [GR86, p. 273–274], Garsia and Remmel noted that the coefficients of the **total $q$-rook number**

$$R(\lambda;q) := \sum_{k\geq 0} R_k(\lambda;q)$$

also appeared to be unimodal. We have verified this for all partitions $\lambda$ such that $|\lambda|\leq 70$. For the staircase $\delta_n$, $R(\delta_n;q)$ is the *$q$-Bell number* [Wag04]. Note that $R(\lambda;q)$ is not necessarily log-concave since $R(\langle 2,1\rangle;q)=1+2q+q^2+q^3$.

**Remark 8.** The $q$-Stirling numbers $S_q(n,k)$ of the second kind, defined by $S_q(n,k) := R_{n-k}(\delta_n; q)$, or alternatively by the recurrence

$$S_q(n,k) = q^{k-1}S_q(n-1,k-1) + [k]_q S_q(n-1,k) \quad \text{for} \quad 0 < k \leq n, \qquad S_q(0,k) = \delta_{0,k}, \tag{5}$$

were further conjectured to be log-concave by Wachs–White [WW91, p. 45]. This conjecture was discussed more recently in work of Sagan and Swanson [SS24, Conj. 7.4], who verified it for $1 \leq k \leq n \leq 50$. We have checked it for $1 \leq k \leq n \leq 250$. See [SS24, §1.2] for a history of these polynomials.

It follows easily from the recurrence (5) that these $q$-Stirling numbers can be obtained from the following evaluation of a *complete symmetric polynomial* [Gou61]:

$$S_q(n,k) = q^{\binom{k}{2}} h_{n-k}([1]_q, [2]_q, \ldots, [k]_q).$$

Such complete symmetric functions are instances of Schur functions: $h_{n-k}(x_1, \ldots, x_k) = s_{n-k}(x_1, \ldots, x_k)$. Could the evaluation above be useful to show log-concavity or unimodality of $S_q(n,k)$, as in the *principal specialization* [Sta99, §7.8] of Schur functions [Sta89, Thm. 13], or by exploiting the *denormalized Lorentzian property* of Schur polynomials [Huh+22]?

**Remark 9.** For all partitions $\lambda$ of size $|\lambda| \leq 75$ and all $k$, the rank-$k$ $q$-rook number $R_k(B_\lambda; q)$ is log-concave whenever $R_1(B_\lambda; q)$ is log-concave (i.e., if the sequence of diagonal lengths $d_i(\lambda)$ are log-concave). Could this be true for all $\lambda$? Note that $R_1(\delta_n; q)$ is log-concave, so this is a strengthening of the Wachs–White conjecture mentioned in the previous remark.

**Remark 10.** In [Sag92, Cor. 2.6], Sagan showed that the $q$-Stirling numbers of the second kind satisfy the following alternative notion of log-concavity:[1]

$$S_q(n,k)^2 - S_q(n,k-1)S_q(n,k+1) \in \mathbb{N}[q].$$

Based on calculations for partitions $\lambda$ such that $|\lambda| \leq 50$, we conjecture that the $q$-rook numbers $R_k(\lambda; q)$ satisfy this notion of log-concavity, i.e., that

$$R_k(\lambda; q)^2 - R_{k-1}(\lambda; q)R_{k+1}(\lambda; q) \in \mathbb{N}[q]$$

for all $k$ and $\lambda$. At $q = 1$, this conjecture is a consequence of [Nij76, Thm. 1]; see [HOW99] for more about log-concavity for the $q = 1$ case.

## Acknowledgements

The second author thanks Jim Haglund for first telling us and Bruce Sagan for reminding us about the question of the unimodality of the Garsia–Remmel $q$-rook numbers. We also thank Jonathan Leake for helpful discussions on Lorentzian polynomials.

The first author thanks Swee Hong Chan, Hong Chen, Guilherme Zeus Dantas e Moura, and Elizabeth Milićević for their scintillating dinner conversation at the Mid-Atlantic Algebra, Geometry, and Combinatorics Workshop (MAAGC 2024). In particular, we thank

[1] See also [Ler90] for an analogous result for the shifted polynomials $q^{-\binom{k}{2}}S_q(n,k)$.

Swee Hong for pointing out (in response to a remark by the first author) that although it is obviously *possible* to compute many thousands or millions of $q$-rook numbers to test the conjecture, it is not clear whether anyone *actually had done so*.

The first author was partially supported by Simons Foundation gift MPS-TSM-00006960. The second author acknowledges the support of the Natural Sciences and Engineering Research Council of Canada (NSERC) funding reference number RGPIN-2024-06246, and was partially supported by the NSF grant DMS-2154019.

Lastly, the authors have no relevant financial or non-financial competing interests to report.

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